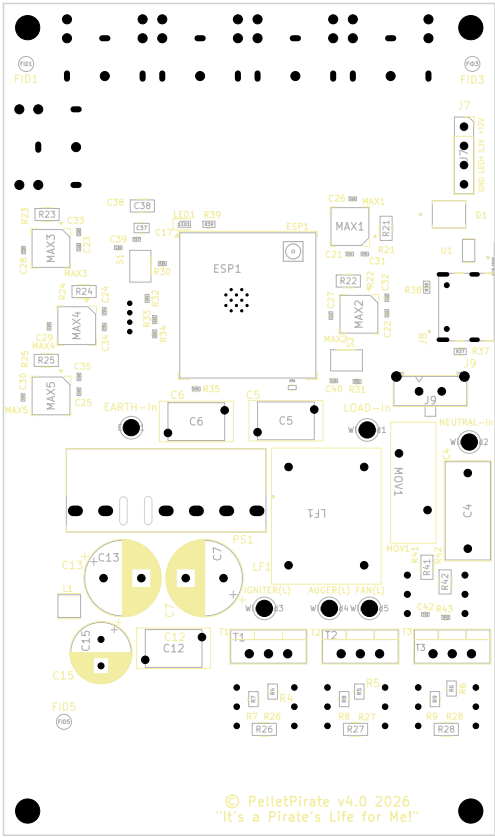
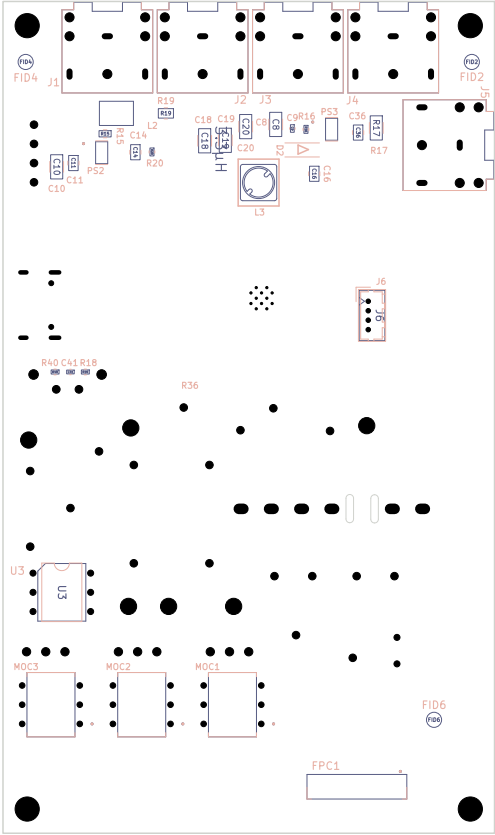


PelletPirate V4 - TOP ASSEMBLY (F.Cu view)



## PelletPirate V4 - BOTTOM ASSEMBLY (B.Cu view, mirrored)



## PelletPirate V4 - SENSITIVE / DO-NOT-SUBSTITUTE (DNS) PARTS

Generated 2026-07-28 | For PCBWay turnkey sourcing

These parts are SAFETY- or FUNCTION-CRITICAL. If PCBWay cannot source the EXACT manufacturer part number below, do NOT accept a substitute - flag it and consign that part instead. Review the parts-match report before paying.

### REF(S): T1, T2, T3

MFR: STMicroelectronics

MPN: BTA12-600BWRG

VALUE: BTA12-600BWRG

DESC: 12A RMS, 600V Off-State Voltage, 50mA Sensitivity, Snubberless, Insulated, Triac, T0-220

LINK: <https://www.digikey.com/en/products/detail/stmicroelectronics/BTA12-600BWRG/654429>

WHY DNS: SNUBBERLESS triac. This exact variant is what makes the on-board snubber deletion safe. A standard (non

### REF(S): MOC3

MFR: ONSEMI

MPN: MOC3053M

VALUE: MOC3053M

DESC: Optoisolator Triac Output 4170Vrms 1 Channel 6-DIP

LINK: <https://www.digikey.com/en/products/detail/onsemi/MOC3053M/9764702>

WHY DNS: RANDOM-PHASE (non-zero-cross) optotriac. REQUIRED for fan phase-angle speed control - a zero-cross subs

### REF(S): MOC1, MOC2

MFR: ONSEMI

MPN: MOC3063M

VALUE: MOC3063M

DESC: Optoisolator Triac Output 4170Vrms 1 Channel 6-DIP

LINK: <https://www.digikey.com/en/products/detail/onsemi/MOC3063M/281237>

WHY DNS: Zero-cross optotriac driver (igniter, auger). Zero-cross switching is required for clean turn-on. Do no

### REF(S): U3

MFR: ON Semiconductor

MPN: H11AA1M

VALUE: H11AA1M

DESC: Optoisolator Transistor with Base Output 4170Vrms 1 Channel 6-DIP

LINK: <https://www.digikey.com/en/products/detail/onsemi/H11AA1M/1053602>

WHY DNS: AC-line zero-cross detector optocoupler. Provides mains->logic isolation for the ZC signal. No substitu

### REF(S): PS1

MFR: Bel Fuse

MPN: PBO-5F-12

VALUE: PBO-5F-12

DESC: Open Frame AC DC Converters 1 Output 12V 420mA 85 ~ 305 VAC, 70 ~ 430 VDC Input

LINK: <https://www.digikey.com/en/products/detail/bel-power-solutions/PBO-5F-12/28717413>

WHY DNS: Isolated AC/DC module - THE primary-to-secondary safety isolation barrier of the whole board. No substi

### REF(S): MOV1

MFR: BOURNS

MPN: MOV-14D561K

VALUE: PP\_14D561K

DESC: 560 V 4.5 kA Varistor 1 Circuit Through Hole Disc 14mm

LINK: <https://www.digikey.com/en/products/detail/bourns-inc/MOV-14D561K/2798251>

WHY DNS: Mains surge varistor. Clamp voltage (560V) is matched to the design. Safety part - no substitution.

### REF(S): C4

MFR: KEMET

**MPN:** R46KF310000M1M  
**VALUE:** 0.1µF 275V X2  
**DESC:** 0.1 µF Film Capacitor 275V 560V Polypropylene (PP), Metallized Radial  
**LINK:** <https://www.digikey.com/en/products/detail/kemet/R46KF310000M1M/5177298>  
**WHY DNS:** X2 safety-rated mains film capacitor (line-to-neutral). Must be certified X2. No substitution.

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**REF(S):** C5,C6  
**MFR:** VISHAY  
**MPN:** VY2471K29Y5SS63V7  
**VALUE:** 470pF 440V X1Y2  
**DESC:** 470 pF ±10% 440VAC Ceramic Capacitor Y5S Radial, Disc  
**LINK:** <https://www.digikey.com/en/products/detail/vishay-beyschlag-draloric-bc-components/VY2471K29Y5SS63V7/19>  
**WHY DNS:** Y1/Y2 safety-rated ceramic capacitors across the isolation barrier (line/neutral to earth). Certified s

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**REF(S):** C12  
**MFR:** VISHAY  
**MPN:** VY2102M29Y5US63V7  
**VALUE:** 1000pF 440V Y5U  
**DESC:** 1000 pF ±20% 440VAC Ceramic Capacitor Y5U (E) Radial, Disc  
**LINK:** <https://www.digikey.com/en/products/detail/vishay-beyschlag-draloric-bc-components/VY2102M29Y5US63V7/19>  
**WHY DNS:** Y-rated ceramic capacitor across the primary-to-secondary barrier. Certified safety part - No substitut

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**REF(S):** C7  
**MFR:** Rubycon  
**MPN:** 450BXG15MEFC10X20  
**VALUE:** 15µF 450V  
**DESC:** 15 µF 450 V Aluminum Electrolytic Capacitors Radial, Can 10000 Hrs @ 105°C  
**LINK:** <https://www.digikey.com/en/products/detail/rubycon/450BXG15MEFC10X20/23330777>  
**WHY DNS:** Primary DC bulk cap. 450V rating is surge headroom tied to the MOV clamp - do NOT drop the voltage rati

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**REF(S):** R41,R42  
**MFR:** Panasonic  
**MPN:** ERJ-P08J153V  
**VALUE:** 15K  
**DESC:** Panasonic ERJ-P08J153V · 15k · 5% 2/3W 1206 anti-surge  
**LINK:** <https://www.digikey.com/en/products/detail/panasonic-industry/ERJ-P08J153V/525339>  
**WHY DNS:** ANTI-SURGE / pulse-rated resistors on the mains zero-cross divider. Must be the pulse-withstanding 2/3W

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**REF(S):** ESP1  
**MFR:** Espressif Systems  
**MPN:** ESP32-S3-WROOM-1U-N16R8  
**VALUE:** ESP32-S3-WROOM-1U-N16R8-flyman  
**DESC:** Bluetooth, WiFi 802.11b/g/n, Bluetooth v5.0 Transceiver Module 2.4GHz Antenna Not Included, U.FL Surfac  
**LINK:** <https://www.digikey.com/en/products/detail/espressif-systems/ESP32-S3-WROOM-1U-N16R8/16162641>  
**WHY DNS:** ESP32-S3-WROOM-1U-N16R8 module (16MB flash / 8MB PSRAM, external-antenna -1U). Exact variant required.

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**REF(S):** MAX1-MAX5  
**MFR:** ANALOG DEVICES  
**MPN:** MAX31865ATP+  
**VALUE:** MAX31865xTP  
**DESC:** Temp Monitoring System (Sensor) External Sensor SPI Output 20-TQFN (5x5)  
**LINK:** <https://www.digikey.com/en/products/detail/analog-devices-inc-maxim-integrated/MAX31865ATP/3681476>  
**WHY DNS:** MAX31865 RTD-to-digital front end. Exact part required for the firmware. No substitution.

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**REF(S):** FPC1  
**MFR:** Hirose Electric Co Ltd  
**MPN:** FH26W-39S-0.3SHW(60)  
**VALUE:** FH26W-39S-0.3SHW\_60\_

DESC: 39 Position FPC Connector Contacts, Bottom 0.012" (0.30mm) Surface Mount, Right Angle  
LINK: <https://www.digikey.com/en/products/detail/hirose-electric-co-ltd/FH26W-39S-0-3SHW-60/5132503>  
**WHY DNS: LCD FPC connector - 0.3mm-pitch staggered, mates the specific LCD ribbon. No substitution.**

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**ALSO NOTE FOR ASSEMBLY:**

- WirePad1-5 and Earth1 are hand-soldered wire lugs: NO machine placement.
- Polarized: C7, C13 (tantalum-polymer), C15 (polymer) - observe + marking.
- PS1 footprint has non-plated routed creepage slots (in the drill/edge data).